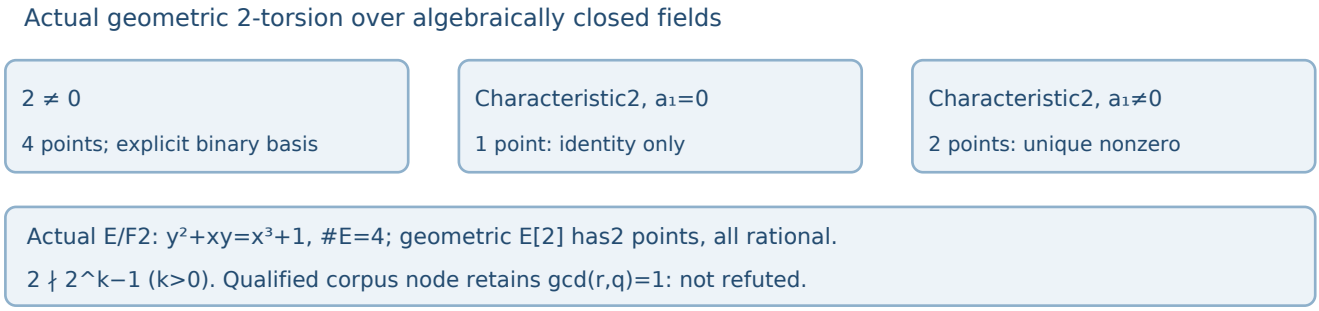
Actual 2-torsion: rank and characteristic boundary

Proved geometry. For every nonsingular general Weierstrass model over a field with $2 \neq 0$, the actual Mathlib subgroup killed by2 is finite and equivalent to identity plus distinct roots of the actual cubic. Its cardinality is $1+\#\text{roots} \leq 4$. Over every algebraically closed field and after every algebraically closed scalar extension it has4points, with an explicit binary basis: two nonzero distinct actual points A/B, full generation0/A/B/A+B, and independence of actual F2coefficient values under the group's scalar action. No rank/basis premise is supplied. Signed standing unverified. [pcc:unverified]



Proof. Actual point negation and the checked coordinate/root construction give the full point-set equivalence. The cubic has discriminant16Δ, so algebraic closedness and $\Delta \neq 0$ give three distinct roots. Finite group/cardinality reasoning with actual multiplicationby2 constructs the basis. Scalar-map injectivity transports $2 \neq 0$ and $\Delta \neq 0$; no extra target-characteristic assumption replaces this step. [pcc:unverified]

Characteristic2. Actual point negation gives $2P=0$ iff $a_1x+a_3=0$ for affine points. When $a_1=0$ and $\Delta \neq 0$, $a_3 \neq 0$, so only identity remains. When $a_1 \neq 0$, a nonzero killed point has $x=-a_3/a_1$ and is unique; algebraic closedness supplies its ordinate through the actual curve equation. The geometric cardinality is1or2. Concrete $y^2+xy=x^3+1$ and $y^2+y=x^3$ both haveΔ1 and cardinalities2/1 over every characteristic2field. [pcc:unverified]

Actual omitted-condition control. For the first curve over F2, an injective coordinate bound5, three distinct actual points and actual order2divisibility force #E=4. Its actual AlgebraicClosure(F2) has two killed-by2points, both images of basepoints under the actual coordinatewise extension. Thus prime2divides #E and its square divides #E, while $2 \nmid 2-1$. Actual binary fields have card2^k and $2 \nmid 2^k-1$ for every positivek. A direct concrete point-inclusion/cyclotomic equivalence negation is proved. [pcc:unverified]

Source check. The full THM-XT-BK-WEIL-SUFFICES node explicitly retains $\gcd(r,q)=1$ and $r \nmid q-1$. Original BK1998Theorem1/proof pp142–143 states a full l^2-point condition; its prime-to-characteristic rank step is invoked after cyclotomic divisibility forces $l \nmid q$. The characteristic2control deletes coprimality only. Neither the qualified node nor the qualified paper is refuted. Exact node/hash and read scope are retained; no corpus status is rewritten. [pcc:unverified]

Checks. Whole gate 67 declarations / 0 failures / 0 trusted native exceptions; Lean4.35.0, Mathlib56a00585…. Complete closed declaration, custom sources/configs and artifact hashes are pinned in sources.json. Failed/successful source/log/exit records and full gate are losslessly archived. Only propext/Classical.choice/Quot.sound; no holes, custom axioms, native evaluator, raised limits, new global elaboration flags or waivers. Exact PCC replay is separate; signed standing unverified. [pcc:unverified]

Remaining scope. Generic geometry at r2 is proved. Generic odd-prime rank/Frobenius determinant/representation/descent, full supersingular geometric/Hasse classification and canonical pairings remain open. No new group-scheme formalization or general Frobenius representation is claimed. Exact existing Mathlib proofs and primary BK were read; unavailable external Silverman is not marked read. KB unavailable, bounded fallback recorded. No novelty, native experiment, paid compute or agent upload. Other all-fourteen obligations remain explicit. [pcc:unverified]